

No Bag Left Behind

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Abstract—Abstract— Unattended bags in public spaces such as airports and train stations pose significant security risks. In this paper, we propose a real-time algorithm for detecting and monitoring unattended bags that integrates advanced computer vision techniques with proximity analysis. Our algorithm employs a Convolutional Neural Network to identify both bags and individuals, while a proximity analysis module evaluates the spatial relationships and distances between them using a defined threshold. Designed to minimize disruptions, the algorithm reduces false positives, ensuring timely and reliable alerts for genuinely unattended bags. The proposed algorithm is rigorously tested across four real-world scenarios, demonstrating exceptional accuracy in identifying unattended bags. Furthermore, it effectively captures information about individuals who left these items behind, providing critical insights for security personnel during investigations. The code and the dataset are publicly available at <https://github.com/ahmadmelebari/No-Bag-Left-Behind.git>.

Index Terms—unattended bag, detection, computer vision, proximity analysis, security, real-time monitoring

I. INTRODUCTION

Unattended bag in public spaces, such as airports and train stations, poses significant security risks. It is a major security concern due to its potential to harbor dangerous materials, and its presence often leads to evacuations and service disruptions. Unattended bag, if left unchecked, can trigger public panic and economic losses for nearby businesses. Figure 1 showcases a sample of unattended bag found on a public area. As public spaces grow more crowded, the detection of unattended bag is becoming more challenging.

The number of unattended bag incidents is expected to increase due to rising public gatherings in transportation hubs, necessitating advanced and reliable detection systems. These incidents demand quick detection to prevent harmful consequences, and traditional surveillance methods relying on human operators often fail to meet the required accuracy and timeliness.

Human-operated surveillance systems struggle in real-world environments, especially in crowded spaces where there are many people and objects moving in video streams. These systems are prone to false positives and false negatives, where unattended bag is either missed or wrongly identified. As a result, there is a pressing need to develop automated systems that can detect and monitor unattended bag with greater accuracy.

Our work is motivated by the success of Convolutional Neural Networks (CNNs) [1, 2, 3, 4] in object detection and real-time image recognition without requiring extensive manual intervention. CNNs have been incorporated into traditional object detection methods, significantly improving the accuracy and reliability of unattended bag detection.

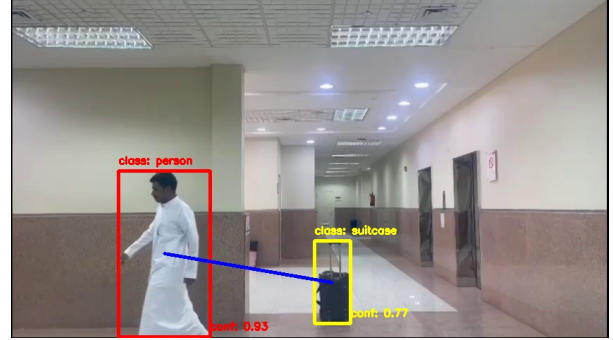


Fig. 1: An example of detecting an unattended bag along with the individual who left it behind using our proposed method.

Additionally, CNNs offer better performance and stability in complex, dynamic environments compared to earlier detection approaches. The main contribution of this paper is as follows:

- 1) Propose a real-time unattended bags and individuals detection algorithm. The algorithm combines CNN-based object detection with proximity analysis to assess the spatial relationship and distance between these objects.
- 2) Demonstrate the efficacy of our algorithm by evaluating its performance across four real-world scenarios.

The rest of this paper is structured as follows: In **Section II** provides a review of existing attempts and datasets used to detect unattended bags. **Section III** presents our proposed detection algorithm. **Section IV** provides experimental results and evaluations. Then, a discussion on limitations and challenges are given in **Section V**. Finally, **Section VI** concludes the paper. Then, it provides future work and potential improvements.

II. RELATED WORK

In this section, we provide related works to this matter. Few research attempts had been published for detecting Unattended bags using video streams. First, Bhargava et al. [5] proposed an intelligent video analysis for unattended object detection in public areas using i-LIDS dataset [6]. Focusing on Unattended bag at train stations, the system traced objects back to their owners and raised alarms if an owner is absent for a predefined period. Then, Ren et al. [7] Utilized a Faster R-CNN using a ResNet101 [8] backbone to detect persons and bags in a single video frame. They built a simple tracker to associate bag and person identities across frames, aiding in abandonment assessment. The system detects a bag as Unattended if it lacks an owner

or if the associated owner is too distant. This dual criterion ensures a comprehensive evaluation. However, their algorithm faced challenges in scenarios with multiple bags and persons, impacting the algorithm's efficiency. Tian et al. [9] introduced a real-time detection of Unattended and removed objects in complex environments through video surveillance. They employed a Gaussian mixture model for background subtraction and static region analysis. Similar to the work in [9], Tian et al. [10] presented an advanced algorithm for reliable detection of unattended and removed objects in complex surveillance videos, using Gaussian mixture models for background modeling and integrating person detection. The system was evaluated on the PETS2006 [11] and i-LIDS datasets with impressive results. Beyan et al. [12] presented a method using background subtraction and thermal images to detect people and unattended bags. Then, Smeureanu et al. [13] used background subtraction and CNNs to detect abandoned bags. Similarly, Altunay et al. [14] employed background subtraction and foreground modeling technique in addition to Faster R-CNNs to detect unattended bags.

Previous studies have made significant advances in the detection of unattended bags in public surveillance systems. However, these approaches have largely overlooked establishing a reliable connection between bags and their owners. In this paper, we introduce a Euclidean distance proximity analysis module to assess spatial relationships and distances between bags and individuals based on a defined threshold. This enables accurate identification of the unattended bag's owner.

III. PROPOSED ALGORITHM

In this section, we show the details of the proposed algorithm. Figure 2 depicts a flowchart of the algorithm for detecting, tracking, and recognizing unattended bags and their associate individuals in a real-time surveillance environment. It begins with data ingestion from surveillance cameras, followed by preprocessing technique to clean and standardize the video data. The Object Detection Module, powered by a pre-trained CNN based detector, YOLO [15], which identifies bags and persons in each frame. The Proximity Analysis function then assesses the spatial relationships between the detected objects to determine potential abandonment.

The algorithm begins by loading YOLO detector for people and bags detection. The YOLO detector is initialized to detect these objects in each video frame in a video stream. The video stream is initialized for frame-by-frame processing. It reads the current frame from the video stream. If the frame is valid, the function continues processing the frame. If the frame is invalid, the function breaks the loop, proceeds to clean up resources, and ends.

The YOLO detector starts to classify and detect people and bags in the current frame. Then, the algorithm checks the Euclidean distance proximity between each detected bag and the detected people. The algorithm iterates over each detected bag to assess its relationship with the closest person using the distance computation and to check its status whether the bag is attended or unattended. The Euclidean distance d is computed as follows:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

where x_1 and y_1 locate the center of a bag bounding box while x_2 and y_2 locate the center a person bounding box.

A threshold distance is predefined. If no person is within the predefined threshold distance, the bag is considered unattended. The algorithm localizes the unattended bag by a yellow bounding box around it. If a person is less or equal the threshold distance, then the algorithm consider the bag as attended and localize the person with a purple bounding box, the bag with a green bounding box, and connects them with a blue line. If the distance exceeds the threshold, the algorithm continues without connecting the distance between the person and bag.

The algorithm further checks the bag unattendancy status for other bags using the distance threshold. If the bag is detected as unattended in previous frames, the algorithm continues to mark the bag with a yellow bounding box in the current frame. If the bag is not marked as unattended in previous frames, the algorithm localizes the bag with a yellow bounding box and the person who left it with a red bounding box. After processing all bag, the algorithm returns the processed frame, which contains the bounding boxes for unattended bag. Then, the returned frame is displayed for the user, showing the bounding boxes and connections between people and bag. The process ends after all frames have been processed, or when no valid frames remain.

IV. EXPERIMENT

In this section, the implementation and the graphical user interface (GUI) of this algorithm are presented. Then, experimental results and evaluation of the fine-tuned model are provided using four real-world scenarios. The scenarios are also described in this section.

A. Implementation

The implementation of our algorithm is divided into two primary components, the core video processing and algorithm script and the GUI. The algorithm is implemented in a Python script, which handles the detection of Unattended bags using the YOLO detector. The script processes each frame of the input video to identify and track objects, specifically focusing on bags and their distance proximity to individuals. The distance proximity function Computes the Euclidean distance between two points to measure proximity and compares bounding boxes across frames to detect stationary or potentially Unattended bags.

The script creates a GUI using Python's using 'tkinter' library [16], as shown in Figure 3. This interface enables users to select video stream sources, either from a connected camera or a video file, adjust model parameters, and initiate the unattended bag detection process.

Users can specify the version of YOLO detector [17] to be used, the device processor (CPU or GPU), the detection confidence, and the distance proximity threshold. The interface facilitates the seamless operation of the video processing system.

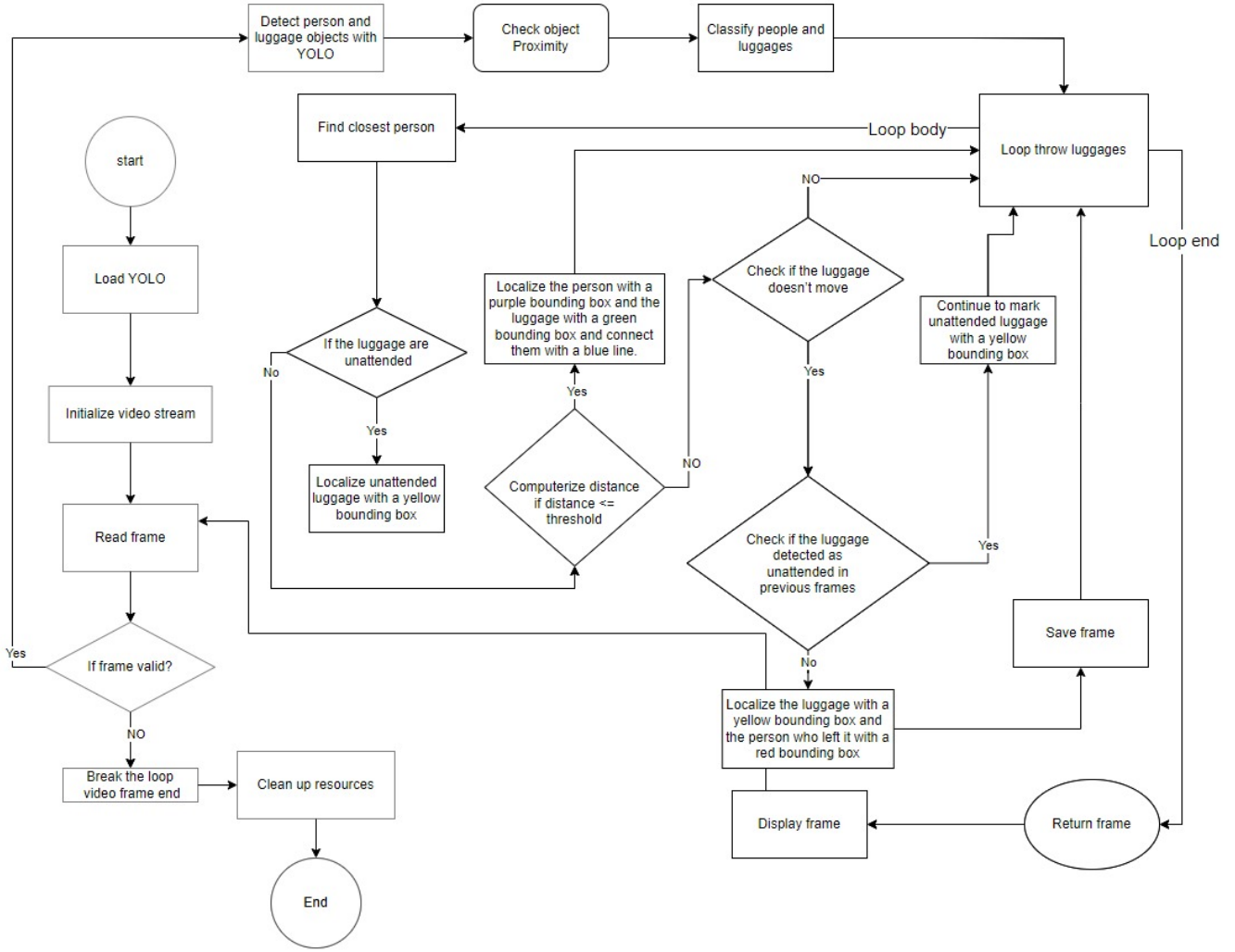


Fig. 2: The proposed algorithm.

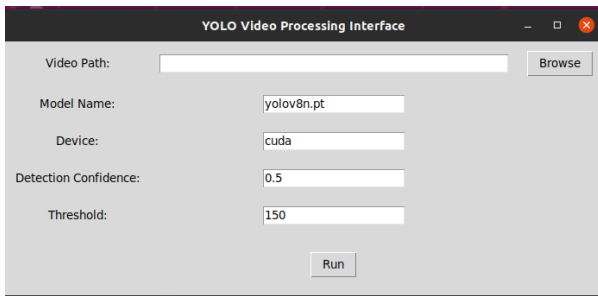


Fig. 3: A user-friendly GUI designed for the proposed algorithm.

B. Description of Four Real-world Scenarios

In this experiment, four different scenarios are used. The scenarios are described as followed:

- **Scenario 1:** In this first scenario, a person arrives carrying a bag, places it down, and then walks away, leaving it unattended. Later, two other individuals pass by the bag without interacting with it. After some time, the bag's owner returns to retrieve it.
- **Scenario 2:** In the second scenario, three people are present: one carrying a bag, another with a backpack,

and a third with a handbag. The individuals with the bag and backpack place their items down and walk away, leaving them unattended. Meanwhile, the person with the handbag passes by the unattended bags without interacting. After some time, the original owners return to retrieve their belongings.

- **Scenario 3:** In the third scenario, a person arrives carrying a bag, leaves it behind, and walks away. Subsequently, another individual arrives with a backpack, places it down, and also walks away. Later, the bag's owner returns to retrieve their bag, followed by the backpack's owner, who comes back to collect their backpack.
- **Scenario 4:** In the fourth scenario, a person carrying a handbag drops it and walks away, leaving it unattended. After some time, the individual returns to retrieve the handbag.

C. Experimental Results and Evaluation

In this section, we present the results of testing our fine-tuned model, based on YOLO version 8, across four distinct scenarios. We use evaluation metrics such as accuracy, precision, recall, and F1 score to assess performance. Additionally, confusion matrices are included to provide a

detailed view of the model's effectiveness in distinguishing between persons and bags.

We first showcase the model's performance for each individual scenario. Following that, we present the final overall performance results across all scenarios.

1) *Performance for scenarios 1:* Table V and Figure 4 show the performance of the model, evaluated using scenario 1. The model performs exceptionally well, achieving an accuracy of 89.69%, precision of 97.75%, recall of 91.58%, and an F1 score of 94.57%. It misclassified 7 instances of bags as persons and 1 instance of a person as a bag. The confusion matrix reveals that the model correctly classifies 40 persons and 51 bags. Misclassifications are represented by the off-diagonal elements, indicating that the model misclassifies 7 bags as persons and 1 person as a bag.

TABLE I: Performance for scenario 1.

Accuracy	Precision	Recall	F1 Score
0.8969	0.9775	0.9158	0.9457

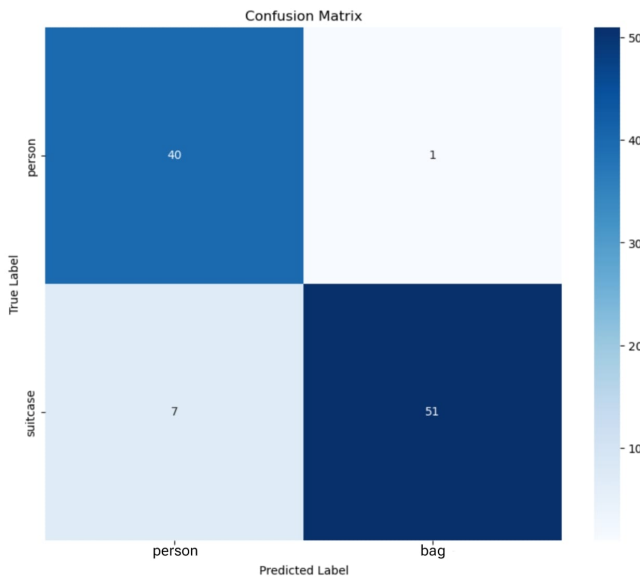


Fig. 4: Confusion Matrix for true versus predicted people and bags categories using scenario 1.

2) *Performance for scenario 2:* Table II and Figure 5 show the performance of the model, evaluated using scenario 2. The model performs moderately, achieving an accuracy of 66.77%, precision of 80.38%, recall of 79.77%, and an F1 score of 80.08%. It misclassifies 54 instances of bags as persons and 46 instances of persons as bags. The confusion matrix indicates that the model correctly classifies 138 persons and 169 bags. However, there are some misclassifications, with 54 bags incorrectly labeled as persons and 46 persons misclassified as bags.

TABLE II: Performance for scenario 2.

Accuracy	Precision	Recall	F1 Score
0.6677	0.8038	0.7977	0.8008

3) *Performance for scenario 3:* Table III and Figure 6 show the performance of the model, evaluated using scenario 3. The model performed moderately, achieving an accuracy

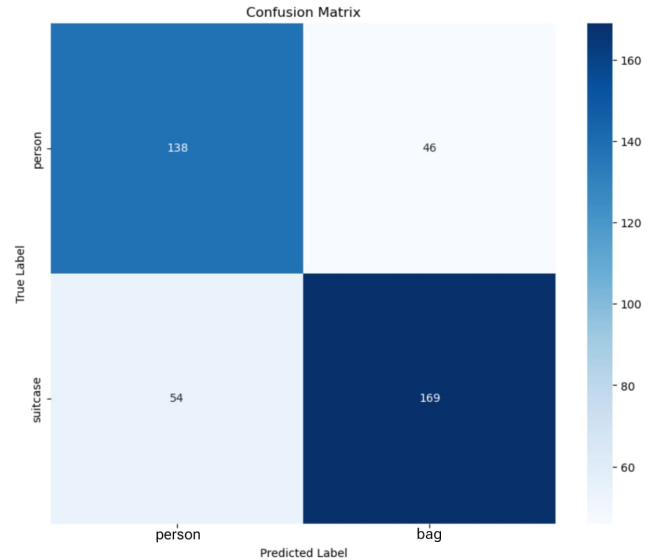


Fig. 5: Confusion Matrix for true versus predicted people and bags categories using scenario 2.

of 71.78%, precision of 95.39%, recall of 74.36%, and an F1 score of 83.57%. It misclassifies 24 instances of bags as persons and 3 instances of persons as bags. The confusion matrix offers a detailed view of the model's classification performance. The diagonal elements indicate correct identifications, with the model accurately detecting 86 persons and 81 bags. Misclassifications are displayed in the off-diagonal elements, revealing instances where the model mistakenly classified some bags as persons and vice versa.

TABLE III: Performance for scenario 3.

Accuracy	Precision	Recall	F1 Score
0.7178	0.9539	0.7436	0.8357

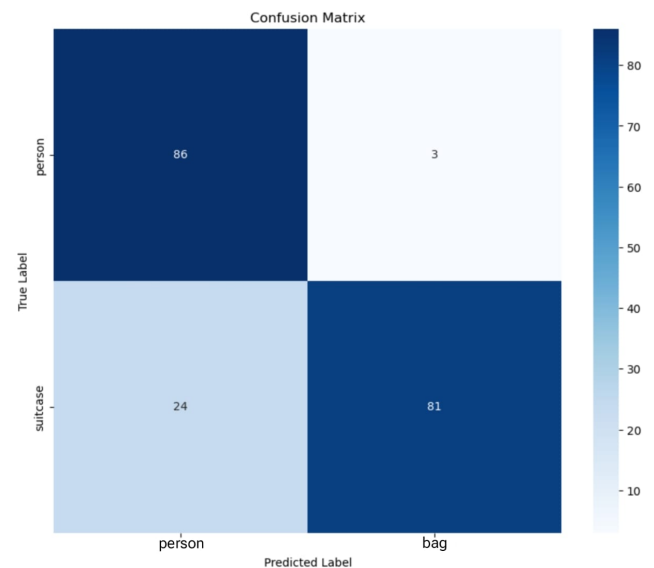


Fig. 6: Confusion Matrix for true versus predicted people and bags categories using scenario 3.

4) *Performance for scenario 4:* Table IV and Figure 7 show the performance of the model, evaluated using scenario 4. The model performs moderately, achieving an accuracy of 60.29%, precision of 100.00%, recall of 60.29%, and an F1 score of 75.23%. It misclassifies 6 instances of bags as persons and 0 instances of persons as bags. The confusion matrix shows that the model correctly identifies 34 persons and 7 bags. However, 6 bags are misclassified as persons, which affected the overall recall.

TABLE IV: Performance for scenario 4.

Accuracy	Precision	Recall	F1 Score
0.6029	1.0000	0.6029	0.7523

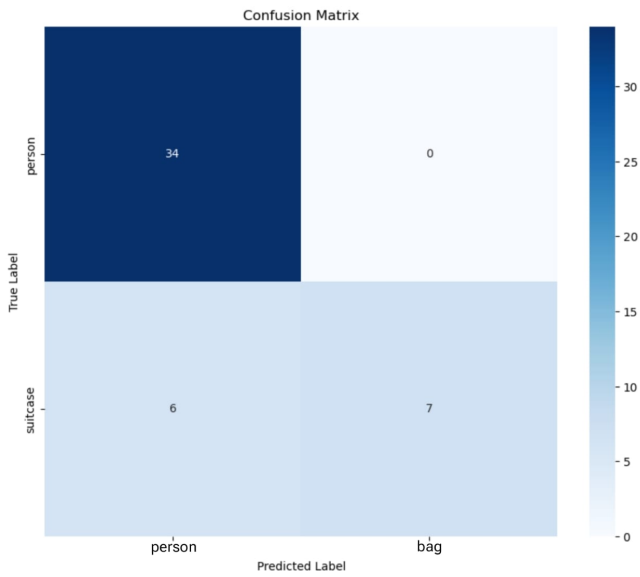


Fig. 7: Confusion Matrix for true versus predicted people and bags categories using scenario 4.

TABLE V: An average performance of the model in detection across scenarios 1, 2, 3, and 4.

Accuracy	Precision	Recall	F1 Score
0.7213	0.9338	0.7650	0.8336

The model performs well across all scenarios, achieving an average detection accuracy of 72.13%, precision of 93.38%, recall of 76.50%, and an F1 score of 83.36%. It shows strong performance in correctly identifying unattended bag while maintaining a good balance between minimizing false positives and false negatives. However, there are instances where some unattended bags are missed, as reflected in the recall value.

The overall results indicate that the fined-tuned model performs exceptionally well across all scenarios in terms of Accuracy, Recall, and F1 Score. The model demonstrates strong generalization capabilities, consistently achieving high performance in detecting and classifying bags and persons.

V. DISCUSSION

The performance of the object detection model is crucial for the overall accuracy and reliability of the proposed

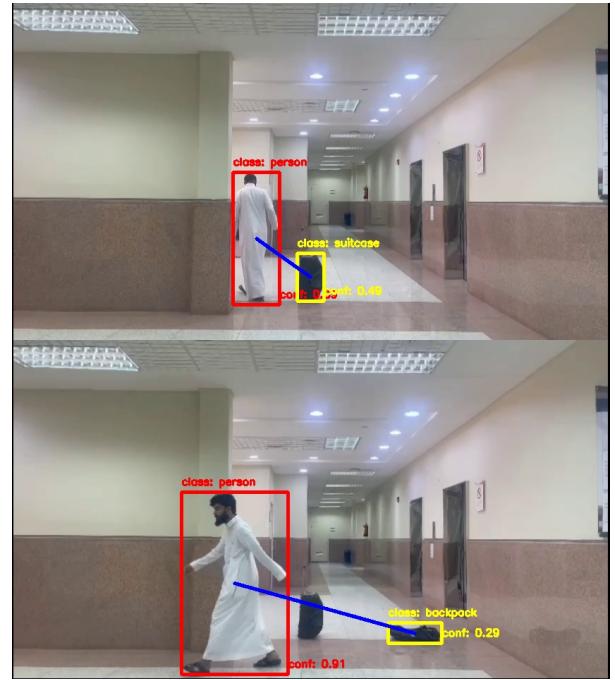


Fig. 8: Qualitative results from different scenarios show bounding boxes around the unattended bags and their respective owners who left them behind, connected by a line.

system. As shown in the experimental results, the fine-tuned model based on YOLOv8x consistently achieves high accuracy, precision, recall, and F1 scores.

One significant challenge in detecting an unattended bag is the model's difficulty in identifying small scale objects, particularly when the bag is lying flat on the floor, as shown in Figure 9. Also, factors such as lighting conditions and camera angle contribute to this limitation, making it more challenging for the model to recognize the bag as a critical object. Furthermore, even when the bag is identified as unattended, the algorithm frequently associates it with the nearest person without confirming whether that individual is the actual owner, as shown in Figure 10. This can result in scenarios where the model incorrectly assumes the bag is attended by a different person while the owner has actually left, posing potential security risks in sensitive areas such as airports or train stations.

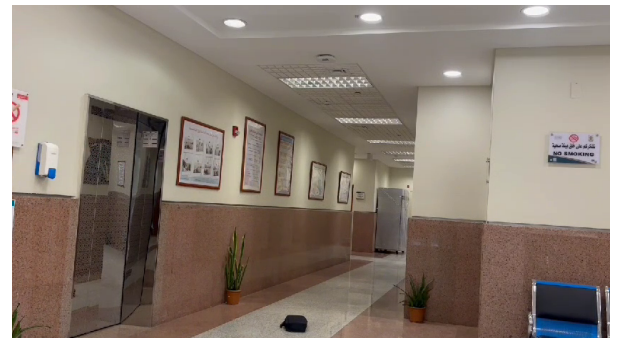


Fig. 9: An example of an unattended bag not detected by our model due to its small scale, low lighting, and unfavorable camera angle.



Fig. 10: An example when the algorithm associates the bag with the closest person, ignoring the real owner.

VI. CONCLUSION AND FUTURE WORK

This paper presents a real-time algorithm for unattended bags detection and monitoring that integrates advanced computer vision techniques with proximity analysis. Our algorithm utilizes a Convolutional Neural Network to detect bags and individuals, while a proximity analysis module evaluates the spatial relationships and distances between them using a defined threshold. The algorithm aims to minimize disruptions by reducing false positives, ensuring timely and reliable alerts for truly unattended bags. Rigorous testing on four real-world scenarios demonstrates the algorithm's exceptional accuracy in identifying unattended bags. Additionally, it effectively captures information about individuals who left these items behind, providing crucial insights for security personnel during investigations.

In future work, we suggest leveraging Re-Identification (ReID) to address the issue of bag ownership verification, thereby enhancing the algorithm's ability to accurately associate bags with their rightful owners. The ReID process should extract distinguishing features of the person initially associated with the bag, such as clothing, colors, or body structure, using deep learning-based models. The method will compare the features of the individual currently near the bag with those of the original owner to determine if this new person is indeed the bag's owner.

REFERENCES

- [1] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [2] Tarik Alafif, Anas Hadi, Manal Allahyani, Bander Alzahrani, Areej Alhothali, Reem Alotaibi, and Ahmed Barnawi. Hybrid classifiers for spatio-temporal abnormal behavior detection, tracking, and recognition in massive hajj crowds. *Electronics*, 12(5):1165, 2023.
- [3] Tarik Alafif, Zeyad Hailat, Melih Aslan, and Xuewen Chen. On detecting partially occluded faces with pose variations. In *2017 14th International Symposium on Pervasive Systems, Algorithms and Networks & 2017 11th International Conference on Frontier of Computer Science and Technology & 2017 Third International Symposium of Creative Computing (ISPAN-FCST-ISCC)*, pages 28–37. IEEE, 2017.
- [4] Mehrez Boulares, Tarik Alafif, and Ahmed Barnawi. Transfer learning benchmark for cardiovascular disease recognition. *IEEE Access*, 8:109475–109491, 2020.
- [5] Medha Bhargava, Chia-Chih Chen, M.S. Ryoo, and J.K. Aggarwal. Detection of abandoned objects in crowded environments. In *2007 IEEE Conference on Advanced Video and Signal Based Surveillance*, pages 528–533. IEEE, 2007.
- [6] i-lids dataset for avss 2007. <ftp://motinas.elec.qmul.ac.uk/pub/iLids>, 2007. Accessed: 2024-09-05.
- [7] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. In *Advances in neural information processing systems*, pages 91–99, 2015.
- [8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, pages 770–778, 2016.
- [9] Ying-Li Tian, Rogerio Feris, and Arun Hampapur. Real-time detection of abandoned and removed objects in complex environments. In *The Eighth International Workshop on Visual Surveillance - VS2008*, Marseille, France, 2008. HAL Id: inria-00325775, Submitted on 30 Sep 2008.
- [10] Yingli Tian, Rogério Schmidt Feris, Haowei Liu, and Arun Hampapur. Robust detection of abandoned and removed objects in complex surveillance videos. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 41(5):565–576, 2011.
- [11] Pets 2006 benchmark data. In *IEEE International Workshop on Performance Evaluation of Tracking and Surveillance (PETS)*, 2006.
- [12] Cigdem Beyan, Ahmet Yigit, and Alptekin Temizel. Fusion of thermal-and visible-band video for abandoned object detection. *Journal of Electronic Imaging*, 20(3):033001–033001, 2011.
- [13] Sorina Smeureanu and Radu Tudor Ionescu. Real-time deep learning method for abandoned luggage detection in video. In *2018 26th European Signal Processing Conference (EUSIPCO)*, pages 1775–1779. IEEE, 2018.
- [14] Damla Gül Altunay, Naciye Karademir, Okan Topçu, and Cem Direkçöglü. Intelligent surveillance system for abandoned luggage. In *2018 26th Signal Processing and Communications Applications Conference (SIU)*, pages 1–4. IEEE, 2018.
- [15] J Redmon. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016.
- [16] Python Software Foundation. *Tkinter: Python's De-facto GUI Framework*, 2024. Accessed: 2024-09-05.
- [17] Ultralytics. *Ultralytics YOLOv8: Real-Time Object Detection*, 2023. Accessed: 2024-09-05.